

Yahoo Japan Corporation: Rapidly developing fraudulent ad screening with the Tech Acceleration Program using AI

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

[About Yahoo Japan Corporation](#)



About Yahoo Japan Corporation

Yahoo Japan Corporation handles a huge number of advertisements, daily. To screen fraudulent advertisements, reaching its view, Yahoo Japan Corporation

Yahoo Japan Corporation is a Japanese internet company originally formed as a joint venture between the American internet company Yahoo Corporation and the Japanese company SoftBank, providing a web portal with media content. In addition to its media business, Yahoo Japan Corporation is now developing a wide range of businesses, such as Yahoo! Auctions and Yahoo! Shopping, plus data solution businesses that utilize huge amounts of big data.

new ad-screening on Google Cloud.

Industries: Media & Entertainment

Location: Japan

Google Cloud results

- Improves efficiency of guideline violation detection by enabling non-engineering teams to create their own models
- Lowers percentage of harmful advertisements served to users
- Enables alignment with Pharmaceutical Machinery Law through machine learning

Yahoo Corp
has
cut
app
9.5 m
fraud
in just one
month

(<https://cloud.google.com/functions/>). Cloud
Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud AutoML (<https://cloud.google.com/automl/>)

Cloud Run (<https://cloud.google.com/run/>) help.

Cloud Functions
(<https://cloud.google.com/functions/>)

Cloud Firestore
(<https://cloud.google.com/firestore/>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

To protect Yahoo Japan Corporation

(<https://about.yahoo.co.jp/en/info/company/>)'s users from malicious adverts, its advertising guidelines prohibit not only adverts that violate the law, but also those that may offend or mislead users. When the company started its own advertising business back in 1996, all adverts were checked manually. But following the rapid growth of ad listings and sponsored content, its team of ad checkers faced challenges in keeping up with the sheer volume of work.

This led to Yahoo Japan Corporation introducing an automated ad screening system to support manual checking. The system was designed to filter out problematic text from advert titles and descriptions, then notify staff of any concerns, but it too began struggling to keep up with the growth in the number of adverts.

"The system has been in operation for a long time and our screening department has been blocking out ads everyday. However, recent years have seen a large increase in the volume of adverts, and the growing sophistication of fraudulent adverts meant that the system was overburdened. We needed to boost our workforce to keep up with the alerts," explains Mr. Ichijo, General Manager of the Trust and Safety Division and the Media Management Division at Yahoo Japan Corporation.

Using AI to identify fraudulent ads

In 2019 Yahoo Japan Corporation decided to build a new ad screening system incorporating AI

technology to overcome these challenges. Mr. Ito, General Manager of the Product Development Department, Trust and Safety Headquarters and Media Management Headquarters at Yahoo Japan Corporation, who led the development of the new system, said that it had to fulfill three requirements: "Firstly, it had to incorporate scalable machine learning, that could support large advert loads and be able to judge tens of millions of requests a day, with less operational overheads. Secondly, it had to be able to store data for the long-term and be able to easily retrieve historical data when required. The previous system could sometimes only store data for three days because it couldn't keep up with the volume it was receiving. Thirdly, the system had to be flexible and able to develop with our needs so that we can quickly and continuously create new machine learning models. This is why our new system consists of several serverless products. Eventually, I want our non-technical team of advert monitors to be able to build their own machine-learning models without having to use code."

Yahoo Japan Corporation considered a number of cloud operators for this new system, but it settled on Google Cloud based on the strength of its scalable agile data warehouse [BigQuery](https://cloud.google.com/bigquery). (<https://cloud.google.com/bigquery>) and the intuitiveness of [Cloud AutoML](https://cloud.google.com/automl) (<https://cloud.google.com/automl>) for training bespoke machine learning models.

"BigQuery is able to integrate seamlessly with our platform, provide unlimited data retention, and excellent search speed, which improves the

efficiency of data analysis, crucial for effective machine learning. Cloud AutoML is also excellent because it's flexible and handles data linkage with ease. It's a product that even non-engineers can use and we requested that it be used in a managed manner," says Ito.

"The system has been in operation for a long time and our screening department has been blocking out ads everyday.

However, recent years have seen a large increase in the volume of adverts, and the growing sophistication of fraudulent adverts meant that the system

—Mr. Ichijo, General
Manager, Trust and Safety
Division and Media
Management Division,
Yahoo Japan Corporation

[illegible]

<https://cloud.google.com/blog/ja/topics/customers/yahoo-tech-acceleration-program>

The diagram above shows how Yahoo Japan Corporation's new ad screening system is configured. The advert data and ad screening results are sent from the existing system. The accumulated data is then put through a machine learning model and the adverts are then examined using AI on Google Cloud.

Ito explains "This mechanism utilizes Cloud Run (<https://cloud.google.com/run>), Cloud AutoML, Vertex AI (<https://cloud.google.com/vertex-ai>), Firestore (<https://cloud.google.com/firestore>) and more from Cloud Functions (<https://cloud.google.com/functions>). Following machine learning preprocessing and prediction, after aggregating predicted values in parallel and judging, it returns the results at high speed, making various predictions."

Yahoo Japan Corporation created the basis of this system with Google's Tech Acceleration Program, a unique program available only in Japan that enables the application modernization team of Google Cloud to build a system that conforms to KPIs in a short period of time. Ordinarily, this process can take over 6 months to complete in house, from application design, to compatibility testing, and environment construction. But with the tech acceleration program, the time was reduced from months to just days. For Yahoo Japan Corporation, this involved a two-day pre-discussion to consider requirements, definition and design, followed by three days of prototyping.

"The most memorable part of the Tech Acceleration Program is the prototyping part," shares Ito. "With

the advice of Google Cloud experts, we've incorporated our business needs and future prospects into Domain Driven Design. It was an invaluable experience. For those of us who have never really used Google Cloud, we really appreciated the generous technical support at the time of implementation."

Yahoo Japan Corporation has now introduced an OK/NG classification machine for adverts that uses a machine learning model based on past performance data. "The OK/NG classifier, created with AutoML Tables, has produced better results than the risk check system, which is a collection of the knowledge and know-how that the review team has accumulated over the years. Although it is not at a stage where it is fully autonomous, it is a huge improvement on our old risk check system, which has reached a plateau," says Ito.

"BigQuery is able
to integrate
seamlessly with
our platform,
provide
unlimited data
retention, and
excellent search

speed, which improves the efficiency of data analysis, crucial for effective machine learning. Cloud AutoML is also excellent because it's flexible and handles data linkage with ease. It's a product that even non-engineers can use and we requested that it be used in a managed manner."

*—Mr. Ito, General
Manager, Product
Development Department,
Trust and Safety
Headquarters, Media
Management*

*Headquarters at Yahoo
Japan Corporation*

"Now, our AutoML Tables can create a model just by inputting the test result and the information used for judgement, so it is easy for non-technical auditors to create, verify, and run a model. We have created and are operating many models other than the OK/NG classifier, and laying a new foundation for making Yahoo Japan Corporation's adverts safer and more secure, the key reason for this development."

To improve the model further additional mechanisms are soon to be added too. Key among these is MLOps. "To improve and maintain the accuracy of the model, we are creating a mechanism for ML Ops that captures new data, checks the captured data for lies, recreates the model, and compares, evaluates, and introduces it," explains Ito.

"We must improve the accuracy of the ad screening system too, to expand the area that can be judged automatically and reduce the area that is difficult to judge by machine and must still be done manually. Ultimately, we will develop the resources to perform the screening itself. Next, we are planning to shift to the work of checking and labeling learning data, creating models, etc. And, although we are now focusing on the advertising area, eventually Yahoo Japan Corporation will be able to sell AI examinations to external partners."

"Google Cloud definitely has potential to be used in other departments of Yahoo Japan Corporation. It will be useful in places where quick wins are especially required and large-scale efforts must be made with a small team of staff. An example would be when a prototype must be created in a hurry due to sudden market needs. Working with Google Cloud makes us confident that we can meet those demands," concludes Ichijo.

